Substituting the values of A and B in (1), we get

$$\frac{7x-1}{x^2-5x+6} = \frac{-13}{(x-2)} + \frac{20}{(x-3)}$$

DO YOU KNOW?

$$\frac{7x-1}{x^2-5x+6} = \frac{1}{(x-2)}\left[\frac{(7x-1)}{(x-3)}\right]_{x=2} + \frac{1}{(x-3)}\left[\frac{(7x-1)}{(x-2)}\right]_{x=3}$$

$$= \frac{-13}{(x-2)} + \frac{20}{(x-3)}$$

Example 2.3

Resolve into partial fraction:

$$\frac{x+4}{(x^2-4)(x+1)}$$

Solution

Write the denominator into the product of linear factors

Here $(x^2-4)(x+1) = (x-2)(x+2)(x+1)$

$$\therefore \frac{x+4}{(x^2-4)(x+1)} = \frac{A}{(x-2)} + \frac{B}{(x+2)} + \frac{C}{(x+1)} \quad \ldots(1)$$

Multiplying both the sides by $(x-2)(x+2)(x+1)$, we get

$$x+4 = A(x+2)(x+1) + B(x-2)(x+1) + C(x-2)(x+2) \quad \ldots(2)$$

Put $x = -2$ in (2), we get

$$-2+4 = A(0) + B(-4)(-1) + C(0)$$

$$\therefore \quad B = \frac{1}{2}$$

Put $x = 2$ in (2), we get

$$2+4 = A(4)(3) + B(0) + C(0)$$

$$\therefore \quad A = \frac{1}{2}$$

Put $x = -1$ in (2), we get

$$-1+4 = A(0) + B(0) + C(-3)(1)$$

$$\therefore \quad C = -1$$

Substituting the values of A, B and C in (1), we get

$$\frac{x+4}{(x^2-4)(x+1)} = \frac{1}{2(x-2)} + \frac{1}{2(x+2)} - \frac{1}{x+1}$$

2.1.2 Denominator contains Linear factors, repeated *n* times

In the rational fraction $\frac{p(x)}{q(x)}$, if $q(x)$ is of the form $(ax+b)^n$ [the linear factor is the product of $(ax+b)$, n times], then $\frac{p(x)}{q(x)}$ can be resolved into partial fraction of the form.

$$\frac{A_1}{(ax+b)} + \frac{A_2}{(ax+b)^2} + \frac{A_3}{(ax+b)^3} + \cdots + \frac{A_n}{(ax+b)^n}$$

For example,

$$\frac{9x+7}{(x+4)(x+1)^2} = \frac{A}{(x+4)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}$$

Example 2.4

Find the values of A, B and C if

$$\frac{x}{(x-1)(x+1)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

Solution

$$\frac{x}{(x-1)(x+1)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

$$x = A(x+1)^2 + B(x-1)(x+1) + C(x-1) \ldots(1)$$

Put $x = 1$ in (1)

$$1 = A(1+1)^2 + B(0) + C(0)$$

$$\therefore A = \frac{1}{4}$$

Put $x = -1$ in (1)

$$-1 = A(0) + B(0) + C(-1-1)$$

$$\therefore C = \frac{1}{2}$$

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